

Dhyan Thakkar

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PROFESSIONAL SUMMARY

Robotics Researcher & Engineer specializing in **Embodied AI**, **Human-Robot Interaction (HRI)**, and **Real-time Kinematic Retargeting** for humanoids and quadrupeds. Demonstrated expertise in developing high-throughput (30 FPS) vision-to-motion pipelines and online system identification for quadrupedal platforms. Recent work: multi-robot humanoid fleet simulation and RL locomotion on the Booster K1 (ROS 2 Jazzy, Isaac Lab), GR00T intent evaluation, and thesis research on MID360 LiDAR simulation with LLM-based planning, ReID, and human tracking.

CORE COMPETENCIES

Embodied AI & Robot Learning

Human-Robot Interaction

Real-time Motion Retargeting (30 FPS)

Reinforcement Learning (PPO, RSL-RL, skr1)

Multi-Robot Fleet Simulation

ROS 2 (Jazzy, Humble) & ROS Noetic

Isaac Sim / Isaac Lab / MuJoCo / Gazebo

LiDAR 3D Detection & Perception

WORK EXPERIENCE

Robotics Gallery, Science City Jan 2024 - Jun 2024

Robotics Intern

Ahmedabad, India

- Developed realtime Visual-Inertial Odometry (VIO) on multicopters, reducing vertical path deviation by 5%.
- Implemented LLM-based navigation using SLAM, leveraging 4D LiDAR and camera-based sensing on Unitree GO2.
- Designed GSLAM-based AMR systems for transporting 200lb+ payloads in warehouse environments.
- Maintained and demonstrated advanced robotic systems (AMRs, quadrupedal robots) for public educational exhibits and technical demonstrations.

Electronics and Communications Department, Nirma University May 2023 - Jul 2023

Inhouse Intern

Ahmedabad, India

- Deployed YOLOv8 with 30 FPS inference achieving 89% accuracy for incoming vehicle classification and 67% precision in speed determination, trained on a custom dataset of 33,000+ images.

Nirma University Sep 2021 - Dec 2022

Research Assistant

Ahmedabad, India

- Streamlined remote development and containerization on multiple HPC systems and clusters.
- Designed a Bi-LSTM model to forecast cloud compute usage, improving resource scaling efficiency.

PROJECTS

Multi-Robot Humanoid Fleet & Locomotion - Booster K1 Aug 2026 - Sep 2026

Built a ROS 2 Jazzy workspace for the 22-DoF Booster K1 humanoid; namespaced multi-robot fleet verified across three simulation backends (Gazebo Harmonic, MuJoCo, Isaac Sim) with SDK-style endpoints. Trained a blind, contact-free velocity locomotion policy (5,000 iterations, RSL-RL) in Isaac Lab 3.0 with teacher-student distillation to a deterministic TorchScript policy. Built a MuJoCo RoboCup 3v3 demo (6 K1s, ball, scripted kick) and a ZED 2i stereo replica in Isaac Sim (± 60 mm baseline, HD720, 101° HFOV, SGBM disparity). Containerized the training stack (Docker + Apptainer) with SLURM-based HPC deployment.

ROS 2 Jazzy, Isaac Sim 6, Isaac Lab 3.0, Gazebo, MuJoCo, RSL-RL

GR00T Intent Evaluation via VQA Jul 2026 - Aug 2026

Built VQA experiment runners to evaluate GR00T model intent predictions on video clips, including clip-to-task matching and a scripted approved-model download pipeline.

PyTorch, NVIDIA Isaac-GR00T

Cosmos3 Video Data Generation Jul 2026

Built a batch video-to-video generation and captioning pipeline (5-second segment captions + event detection via Cosmos3 Reasoner) with segmentation, style/scene variation generation, and Google Drive sync.

Python, Cosmos3, ffmpeg, rclone

SO-101 Manipulation RL Environments 2026

Developed single- and dual-arm SO-101 environments (6/12 joints, 5-6 cameras) spanning seven manipulation tasks (pick-lift, pick-and-place, push-T, ramp/bridge pushes, dual-arm cylinder grasp/reach), with a backend-agnostic RL training layer (skrl / rsl_rl) plus teleop data collection.

MuJoCo, Isaac Sim, skrl, RSL-RL

Thesis - MID360 LiDAR Simulation 2026

Sim work for Livox MID360 LiDAR-based perception: simulated LiDAR capture and 3D detection pipeline setup (LivoxDetection / OpenPCDet) for the thesis platform.

Isaac Sim, Livox MID360, OpenPCDet

Thesis - LLM as Planner with ReID & Human Tracking 2026

LLM-based planner pipeline with person re-identification and multi-object tracking for human action understanding and human character identification, using VoxelNeX for LiDAR detection and LiDAR-HMR for human mesh recovery.

Python, VoxelNeX, LiDAR-HMR, LLMs

Real-time Human-to-Robot Motion Retargeting (GVHMR + GMR) Oct 2025

Engineered a low-latency pipeline utilizing Gravity-Constrained Mesh Reconstruction to recover 3D human sequences from monocular RGB feeds. Implemented Gaussian Mixture Regression (GMR) and TWIST for motion mapping, achieving stable 30 FPS on limited hardware (6GB VRAM). Benchmarked against the AMASS dataset for high-fidelity motion mimicking on humanoid platforms in physics-based simulations.

Isaac Sim, Mujoco, ROS

Online System Identification for Quadrupedal Pushing/Pulling Dec 2025

Adaptive framework for quadrupeds to perform online estimation of object properties (Mass, CoM, Moments of Inertia). Achieved 89% success rate on goal-oriented manipulation tasks with mean angular error < 10 degrees. Optimized full-body control strategies for dynamic interactions with unknown payloads in real-time environments.

ANYmal, Unitree Go1

Anthropomorphic Trajectory Optimization Jan 2026

Investigated robot arm trajectory parameterization to align with natural human kinematic constraints and bell-shaped velocity profiles. Developed velocity metrics for reaching tasks to improve predictability and safety in shared human-robot environments.

Python, Kinematic Constraints

Visual Tokens based Navigation Apr 2025

Collaborated with 4-person team to engineer a navigation system using QWEN2.5VL 8B with scene graphs. Fine-tuned Qwen2.5 and InternVL3 for visual scene grounding, achieving 78.64% on ObjectNav and 62.35% on OVMM. Boosted landmark pose estimation by 27% using camera intrinsics as tokens to hybrid VLM model.

Python, VLMs, Hugging Face, Scene Understanding

Object Centric RL Based UAV Control Nov 2024

Reduced VIO jumps by 80% using RL-based UAV control with RGB-D & EKF pose estimation and segmentation-based object detection. Trained PPO agent for hovering and orbiting a targeted object inside Isaac Sim/Lab.

Python, NVIDIA Isaac Sim, PX4, OpenCV

LLM-based Navigation with LiDAR on Unitree Go2 Apr 2024

Deployed conversational agent on Unitree Go2 quadruped with 15+ retrievable goal landmark tokens for navigating a 1700+ sq.ft. dynamic indoor environment with real-time path planning and localization. Achieved sub-782ms latency on custom audio pipeline for edge-deployed finetuned LLM.

Python, OpenCV, ChatGPT, Unitree SDK2

vSLAM on Locobot for Room Navigation and Maze Solving Dec 2023

Implemented backtracking-based vSLAM systems for mapping 500+ sq.ft. indoor spaces and solving dynamic mazes. Reduced manipulation time by 20% via dynamic dense point cloud segmentation based on color and size.
ROS-Noetic, Intel RealSense

Smart Road Assistance for Visually Impaired v1 Jul 2023

Achieved 89% accuracy on YOLOv8 models trained on 33,000+ image dataset for traffic scene prediction. Reduced crash probability by 30% vs. first-order baseline estimator using custom 2nd-order vehicle motion model with sub-500ms latency.
Jetson Nano, PyTorch, YOLO, OpenCV

EDUCATION

MS in Robotics University of Minnesota - Twin Cities	Aug 2024 - Dec 2026
Minneapolis, MN	
Bachelor's in Electronics and Communications Institute of Technology, Nirma University	Sep 2020 - May 2024
Ahmedabad, India	

AWARDS & HONORS

CR2: Contact-Rich Quadrupedal Pushing	ICRA 2026 Workshop on Contact-Rich Manipulation	2026
MVMS: RNN based Pro-Active Resource Scaling in Cloud Environment	Publication	2023
Outreach Volunteer & Summer Instructor	Minnesota Robotics Institute (MnRI)	2024 - Present
Student Representative, Whole Body Control Technical Committee	IEEE Robotics and Automation Society	Present
Vice Chairperson	IEEE Student Branch Nirma University (SBNU)	2023 - 2024

SKILLS

Languages: Python, C/C++, CUDA, Shell, MATLAB, Docker
Simulation & Physics: NVIDIA Isaac Sim/Lab, MuJoCo, Gazebo, Habitat Sim, AWS Robomaker
Robotics Frameworks: ROS 2 (Jazzy, Humble), ROS (Noetic), Nav2 Stack, Unitree SDK2, PX4, MAVROS
Machine Learning: PyTorch, Hugging Face, RLLib, RSL-RL, skrl, Deepspeed, MMCV, YOLOv8, Bi-LSTM
Hardware & Systems: Jetson Nano, Raspberry Pi, Unitree Go1/Go2/G1, Booster K1, SO-101, ANYmal, Agate HPC Cluster
Computer Vision: RGB-D Sensing, LiDAR 3D Detection (Livox MID360, VoxelNeX, OpenPCDet), EKF Pose Estimation, vSLAM, G-SLAM, Intel RealSense
Datasets: AMASS, nuScenes3D, Waymo Open, KITTI, Habitat-Matterport 3D, HSSD, RxR